XISOSIS · Institutional Research Paper

$\mathbb{Z}/16$ BORDISM CHARACTERS AND THE DETECTION OF THE CAPPELL–SHANESON EXOTIC $\mathbb{R}P^4$

Main result: $\mathbb{Z}/16$ bordism characters define an invertible four-dimensional super-TQFT whose closed values, sixteenth roots of unity, separate $\mathbb{R}P^4$ from the Cappell–Shaneson exotic $\mathbb{R}P^4$ for every pairing of tangential structures; no theory of this kind can detect a homotopy 4-sphere.

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Abstract. We give a detailed lemma-and-proof solution of Exercise 9, the exercise explicitly linked to Problem 5 in the supplied announcement. We compute the stable tangent bundle and Stiefel–Whitney classes of $\mathbb{R}P^4$, prove non-orientability, prove existence of precisely two Pin^+ structures, and exclude Pin^- structures; by homotopy invariance of Stiefel–Whitney classes the same conclusions hold for every smooth manifold homotopy equivalent to $\mathbb{R}P^4$, including the Cappell–Shaneson exotic projective space. We then prove an explicit separation theorem using the known Pin^+ bordism classes of standard and Cappell–Shaneson projective space, and show that the resulting invertible theory cannot see homotopy 4-spheres. We distinguish this theorem from a complete construction satisfying every clause of Problem 5. In particular, "finite categorical data" and the required antilinear action need a precise model. The appendices reproduce the exact motivating exercise, explain the passage from that exercise to the research question, and record the lemma-by-lemma assessment of this edition.

Status. This edition was re-assessed lemma by lemma on 3 November 2026 (Appendix C). The elementary calculations are correct; several proofs were tightened, two citations were corrected, one missing lemma (Lemma 2.7) and one new limitation (Lemma 4.5) were added, and the disposition of the first entry point was corrected. Theorem 3.6 is a consequence of established external results. The paper is not a complete, self-contained solution to Problem 5. Deep bordism and field-theory results are explicitly identified as external inputs; their proofs are not reproduced. No independent specialist or proof-assistant verification is claimed. Prepared with OpenAI ChatGPT; reviewed and corrected with Claude (Anthropic), which checked each proof and the cited sources it could access, as recorded in Appendix C.

Source attribution: Roger Cooperman and A.E. Petrunina V.V., *The Cooperman–Petrunina Challenge Series: Announcement, Primer and Reading Guide* (2026), supplied PDF, Part I, pp. 8–10 and 13. Quoted source text is attributed under its CC BY 4.0 licence.

1 The exact task and the scope of the answer

The attached file contains an announcement/primer and the underlying algebra paper. Its Part I, page 8, asks:

> Is there a four-dimensional TQFT, built from finite (possibly non-semisimple, non-unitary) categorical data, defined on Pin manifolds and carrying an antilinear involution T with $T^2 = (-1)^F$, that distinguishes two closed smooth 4-manifolds which are homeomorphic but not diffeomorphic?

The primer explicitly names Exercise 9 in its Problem 5 reading path. There are no separately numbered Problem 5 exercises beyond that one in the supplied file. Exercises 1 and 5 provide general quaternionic background; Exercise 10 concerns Problem 1 but explains the shared antiunitary mechanism. Their solutions are included as supplementary material in Section 5. Exercises 2–4 and 6–8 are not identified as Problem 5 exercises and are not relabelled as such here.

The three bullets called "Entry points" are research questions, not solved exercises. They concern extending the obstruction theorem to Pin theories, distinguishing the Cappell–Shaneson example, and defining finite categorical data. Sections 3–4 address what is established and what is still missing for these requests.

The announcement says that the separate five-problem deposit is authoritative if its formulation differs. That document is not part of the attached Part II, which is the underlying quantum-information paper. Its DOI could not be accessed during this review. Accordingly this paper addresses the statement actually supplied; it does not certify compliance with unseen additional hypotheses.

Conventions and foundations

Cohomology has coefficients in $\mathbb{F}_2$ unless indicated otherwise. ε is a trivial real line bundle, γ the tautological line bundle on real projective space, and $a = w_1(\gamma)$. Pin structures are lifts of the orthonormal frame bundle of the *tangent* bundle. All manifolds considered here are smooth. Fermion parity is the grading operator $P = (-1)^F$ on a complex super vector space.

We use the basic definitions and properties of characteristic classes, Clifford algebras, cellular cohomology, and principal bundles. Calculations and inferences retain their lemma-and-proof presentation. Lemmas 4.2–4.4 and A.3–A.5 include observations about the scope of this argument; they are not general non-existence theorems. This does not mean reproving all foundational topology from set theory. The specifically deep inputs used in Section 3 are separated from the calculations, rather than disguised as elementary lemmas.

Lemma 1.1 (The wording "involution"). On a space with a nonzero odd sector, a map satisfying $T^2 = P$ is not an involution in the literal sense $T^2 = I$.

Proof. For an odd vector $v \neq 0$, $Pv = -v$. Thus $T^2 v = -v \neq v$. If both equalities were required, the odd sector would vanish. Consequently the intended terminology must be a time-reversal symmetry whose square is parity, or an involution modulo parity. This linguistic repair alone neither constructs nor obstructs the requested field theory. □

2 Exercise 9: complete solution

Lemma 2.1 (Cohomology and the tautological class). $H^*\left(\mathbb{RP}^4; \mathbb{F}_2\right) = \mathbb{F}_2[a]/\left(a^5\right)$, with $\deg a = 1$ and $a = w_1(\gamma)$. In particular a and a^2 are nonzero.

Proof. The usual CW structure on $\mathbb{RP}^4$ has one cell in each dimension $0, 1, 2, 3, 4$. Each integral cellular boundary is multiplication by either 0 or 2, so all cellular boundaries vanish modulo 2. There is therefore one copy of $\mathbb{F}_2$ in each of these cohomological degrees and no groups above degree four.

For completeness the cup products can be read from $\mathbb{RP}^\infty = B(\mathbb{Z}/2)$. Let g generate $\mathbb{Z}/2$. In the normalized bar complex with trivial $\mathbb{F}_2$ coefficients, the degree-one cocycle u has $u(g) = 1$. Its k-fold cup power takes value 1 on $[g|\cdots|g]$. These bar chains are cycles modulo two: interior terms contain $g^2 = 1$ and vanish after normalization, and the two endpoint terms cancel. Thus u^k is not a coboundary. Since there is one cohomology generator in every degree, the ring is $\mathbb{F}_2[u]$. Restriction to the four-skeleton preserves these classes through degree four and kills degrees above four, giving the asserted truncated polynomial ring.

On $\mathbb{RP}^1$, the tautological line bundle is the Möbius line bundle: lifting a generator loop reverses the sign of its real fibre. Its first Stiefel–Whitney class is nonzero. Hence $w_1(\gamma)$ is exactly the nonzero degree-one generator, which we denote by a. □

Lemma 2.2 (Stable tangent bundle). $T\mathbb{RP}^4 \oplus \varepsilon \cong 5\gamma$ as real vector bundles.

Proof. At a line $\ell \subset \mathbb{R}^5$, a first-order change of the line is a linear map $\ell \to \mathbb{R}^5/\ell$. Thus $T\mathbb{RP}^4 \cong Hom\left(\gamma, \varepsilon^5/\gamma\right)$. Choose a Euclidean metric so the quotient identifies with $\gamma^\perp$. Then

$$Hom\left(\gamma, \gamma^\perp\right) \oplus Hom(\gamma, \gamma) \cong Hom\left(\gamma, \varepsilon^5\right).$$

The summand $Hom(\gamma, \gamma)$ is canonically trivial, being generated by the identity endomorphism. A metric identifies γ^* with γ, and the right-hand side is $5\gamma^* \cong 5\gamma$. Adding the trivial line $\varepsilon \cong Hom(\gamma, \gamma)$ to $T\mathbb{R}P^4 \cong Hom(\gamma, \gamma^\perp)$ therefore gives $T\mathbb{R}P^4 \oplus \varepsilon \cong 5\gamma$. □

Lemma 2.3 (Characteristic-class calculation). The total Stiefel–Whitney class of $\mathbb{RP}^4$ is $1 + a + a^4$. In particular $w_1 = a$, $w_2 = 0$, $w_3 = 0$, and $w_4 = a^4$.

Proof. Stability and the Whitney product formula applied to Lemma 2.2 give

$$w\left(T\mathbb{RP}^4\right) = w(5\gamma) = (1+a)^5.$$

Expanding gives $1 + 5a + 10a^2 + 10a^3 + 5a^4 + a^5$. In $\mathbb{F}_2$, the coefficients reduce to $1, 1, 0, 0, 1, 1$, respectively; Lemma 2.1 gives $a^5 = 0$. Reading off homogeneous components yields the result. □

Lemma 2.4 (Why the two Pin obstructions differ). A real vector bundle E admits a Pin$^+$ lift if and only if $w_2(E) = 0$, and a Pin$^-$ lift if and only if $w_2(E) + w_1(E)^2 = 0$.

Proof. For either central extension $1 \to \{\pm 1\} \to Pin^{\pm}(r) \to O(r) \to 1$, lift transition functions on a good cover. Their failure to obey the cocycle identity on triple intersections is a $\{\pm 1\}$-valued 2-cocycle. Changing the lifts changes it by a coboundary. Its vanishing is necessary and sufficient: a correcting 1-cochain then makes the lifted transitions an actual cocycle. There are no further obstructions to this lift through a discrete central double cover.

To identify this degree-two obstruction, note first that it is natural: for a bundle classified by $f : B \to BO(r)$ it equals $f^*(o_\pm)$, where $o_\pm \in H^2(BO(r); \mathbb{F}_2)$ is the class of the central extension $Pin^{\pm}(r) \to O(r)$ (the Čech cocycle above is the pullback of that extension cocycle). It therefore suffices to identify $o_\pm$. Restriction to the diagonal sign-change subgroup $D = (\mathbb{Z}/2)^r \subset O(r)$ is injective, $H^*(BO(r); \mathbb{F}_2) \to H^*(BD; \mathbb{F}_2) = \mathbb{F}_2[x_1, \dots, x_r]$, with image the symmetric polynomials and $w_k \mapsto e_k(x_1, \dots, x_r)$; this is the classifying-space form of the splitting principle. To compute the restricted extension class, choose Clifford generators e_i with $e_i\, e_j = -e_j\, e_i$ for $i \neq j$ and $e_i^2 = (-1)^\epsilon$, where $\epsilon = 0$ for Pin$^+$ and $\epsilon = 1$ for Pin$^-$ (the convention of Kirby–Taylor [8], in which a lift of a reflection squares to $+1$ in Pin$^+$). The preimage of D in $Pin^{\pm}(r)$ is generated by $\pm e_1, \dots, \pm e_r$. For $g = (g_1, \dots, g_r)$ choose the lift $s(g) = e_1^{g_1} \cdots e_r{}^{g_r}$. Moving each factor of the second word leftward past the later factors of the first, and then combining equal generators, gives

$$s(g)s(h) = (-1)^{c(g,h)} s(g+h), \qquad c(g,h) = \sum_{i>j} g_i h_j + \epsilon \sum_i g_i h_i \ (mod\ 2).$$

The coordinate homomorphism $g \mapsto g_i$ represents x_i. Hence the extension cocycle represents

$$\sum_{i>j} x_i x_j + \epsilon \sum_i x_i^2 = w_2(E) + \epsilon w_1(E)^2.$$

Here the identification uses $w_2 \mapsto e_2(x) = \Sigma_{i>j}\, x_i\, x_j$ and $w_1 \mapsto e_1(x)$, together with $(\Sigma_i\, x_i)^2 = \Sigma_i\, x_i^2$ in characteristic two. Injectivity of restriction to BD gives $o_\pm = w_2 + \epsilon w_1^2$ in $H^2(BO(r); \mathbb{F}_2)$, and naturality gives the obstruction $w_2(E) + \epsilon w_1(E)^2$ on the original base. Its vanishing is precisely the lifting criterion established above, proving both assertions. □

Lemma 2.5 (All three conclusions of Exercise 9). $\mathbb{RP}^4$ is non-orientable, admits Pin$^+$ structures, and admits no Pin$^-$ structure.

Proof. A bundle is orientable exactly when its determinant line is trivial, equivalently when its first Stiefel–Whitney class vanishes. Lemmas 2.1 and 2.3 give $w_1 = a \neq 0$, so $\mathbb{RP}^4$ is non-orientable. Its Pin$^+$ obstruction is $w_2 = 0$, so Lemma 2.4 supplies a lift. Its Pin$^-$ obstruction is $w_2 + w_1^2 = a^2 \neq 0$, so no such lift exists. Each part of the exercise follows separately. □

Lemma 2.6 (Number of Pin$^+$ structures). There are precisely two isomorphism classes of Pin$^+$ lifts of the tangent bundle of $\mathbb{RP}^4$, with the base and frame bundle fixed.

Proof. Fix one lift, whose existence is Lemma 2.5. The ratio of any other system of lifted transition functions to this one is a central $\{\pm 1\}$-valued 1-cocycle. Two such systems are isomorphic over the frame bundle precisely when their ratio is a coboundary. The set of lifts is therefore a torsor for $H^1\left(\mathbb{RP}^4; \mathbb{F}_2\right) = \mathbb{Z}/2$. A torsor has the same cardinality as its acting group, hence two elements. This counts lifts over a fixed manifold, not orbits under all diffeomorphisms. □

Lemma 2.7 (The same conclusions for any homotopy $\mathbb{R}P^4$). Let M be a closed smooth 4-manifold homotopy equivalent to $\mathbb{R}P^4$. Then $w(TM) = 1 + a + a^4$; M is non-orientable, admits no Pin^- structure, and admits exactly two isomorphism classes of Pin^+ structures.

Proof. By Wu's formula the Stiefel–Whitney classes of the tangent bundle of a closed manifold are determined by the Steenrod squares on its mod-2 cohomology and Poincaré duality, so they are homotopy invariants: a homotopy equivalence $h : M \to \mathbb{R}P^4$ satisfies $h^* w(T\mathbb{R}P^4) = w(TM)$. Lemma 2.3 then gives $w_1(M) = a \neq 0$, $w_2(M) = 0$ and $w_2(M) + w_1(M)^2 = a^2 \neq 0$, and Lemma 2.4 gives the existence and non-existence statements. Since $H^1(M; \mathbb{F}_2) \cong H^1(\mathbb{R}P^4; \mathbb{F}_2) = \mathbb{Z}/2$, the torsor argument of Lemma 2.6 gives exactly two Pin^+ structures. □

3 A precise detection theorem

Let $R = \mathbb{RP}^4$ and let Q be the Cappell–Shaneson exotic projective 4-space. Do not confuse this Q with the source's quaternion algebra $\mathbb{Q}$.

External inputs, not proofs supplied here

We use the following established results. They are essential dependencies and are *not* claimed to follow from Exercise 9.

Input B (bordism and the exotic pair). (i) Q is homeomorphic to R (Hambleton–Kreck–Teichner [9]; see also [3, §1.1]) but not diffeomorphic to it (Cappell–Shaneson [10]). (ii) $\Omega_4^{Pin^+} \cong \mathbb{Z}/16$, generated by $\mathbb{R}P^4$ (Kirby–Taylor [8]). (iii) Under a suitable identification $\Omega_4^{Pin^+} \cong \mathbb{Z}/16$, the two Pin^+ structures of R represent ± 1 and the two Pin^+ structures of Q (Lemma 2.7) represent ± 9; this is due to Stolz's η-invariant computation [6], in the form stated by Debray [7, Example 3.4]. The η-invariant of the Pin^+ Dirac operator on Q is also computed by Oledzki [4].

Input E (extension to a functor). Over $\mathbb{C}$, invertible super field theories with tangential structure ξ are uniquely determined by, and arise from, characters $\omega : {\Omega_d}^{T\xi} \to \mathbb{C}^\times$ of the semistable tangential bordism group, their closed values being the character values [3, Theorem 3.14 in arXiv v3; the draft of this paper cited "Theorem 3.15", which does not match that version]. Composing stable bordism characters with the natural stabilization map supplies such characters.

These inputs are cited results, not a self-contained derivation of the exotic manifold or of the classification theorem. A literal proof of every step including these results would require substantial further work.

Lemma 3.1 (An explicit finite character). Let $\zeta = e^{2\pi i/16}$. The assignment $\chi(k) = \zeta^k$ is a well-defined injective homomorphism $\mathbb{Z}/16 \to \mathbb{C}^\times$.

Proof. If $k' = k + 16m$, then $\zeta^{k'} = \zeta^k \left(\zeta^{16}\right)^m = \zeta^k$, proving well-definedness. The identity $\zeta^{k+l} = \zeta^k \zeta^l$ proves multiplicativity. Finally $\zeta^k = 1$ holds precisely when 16 divides k, since ζ is a primitive sixteenth root. Thus the kernel is zero. □

Lemma 3.2 (The closed-manifold invariant). Using Input B, set $I(M,p) = \chi([M,p])$ for a closed Pin$^+$ four-manifold. This is a diffeomorphism invariant of structured manifolds, multiplicative under disjoint union, and trivial on Pin$^+$ boundaries.

Proof. A structured diffeomorphism identifies bordism classes, and disjoint union is the addition law in the bordism group. Lemma 3.1 converts that addition into multiplication. A boundary represents zero, and $\chi(0) = 1$. Each assertion follows directly from the definition of a bordism group and a group character. □

Lemma 3.3 (Separation independent of labelling). No choice of Pin$^+$ structure on R gives the same value of I as any choice on Q.

Proof. Input B and Lemma 3.1 give the two sets of values

$$\mathcal{J}(R) = \{\zeta, \zeta^{-1}\}, \qquad \mathcal{J}(Q) = \{\zeta^9, \zeta^{-9}\}.$$

The corresponding exponent sets modulo 16 are $\{1, 15\}$ and $\{9, 7\}$. They are disjoint, so injectivity of χ gives disjoint value sets. A diffeomorphism of underlying smooth manifolds would pull back a Pin$^+$ structure and preserve its value, contradicting this disjointness. □

Lemma 3.4 (A real unstructured numerical separation). For these two manifolds, the averages over their Pin$^+$ lifts satisfy

$$A(R) = \cos(\pi/8), \quad A(Q) = -\cos(\pi/8)$$

for the identification of Input B and the character χ of Lemma 3.1. For any other identification or any other injective character the values become $\cos(\pi u/8)$ and $-\cos(\pi u/8)$ for some odd u; they are unequal for every such choice and independent of any labelling of the lifts.

Proof. Define $A(M) = \frac{1}{2}\Sigma_p I(M,p)$ for $M = R,\ Q$, each of which has two lifts (Lemmas 2.6 and 2.7). Then $A(R) = (\zeta + \zeta^{-1})/2 = \cos(\pi/8)$. Since $\zeta^8 = -1$, one has $\zeta^9 = -\zeta$ and $\zeta^{-9} = -\zeta^{-1}$, proving the formula for Q. Changing the identification or the injective character replaces ζ by ζ^u with u odd, and the same computation gives $\pm\cos(\pi u/8)$; $\cos(\pi u/8) \neq 0$ because $\pi u/8$ is never an odd multiple of $\pi/2$ when u is odd. Permuting the two lifts leaves each sum unchanged. This is a numerical invariant calculation; it does not by itself prove a gluing theorem for a TQFT. □

Lemma 3.5 (Functorial extension, with dependency exposed). Using the bordism identification in Input B and the classification theorem in Input E, the invariant I is the closed-manifold partition function of an invertible four-dimensional Pin$^+$ super TQFT.

Proof. Take the unstable tangential structure $\beta : BPin^+(4) \to BO(4)$. Its semistable bordism group $\Omega_4^{T\beta}$ retains a lift to $BPin^+(4)$ after stabilization of the tangent bundle. The map $BPin^+(4) \to BPin^+$ forgets this rank-four refinement and induces the homomorphism

$$\Omega_4^{T\beta} \xrightarrow{s} \Omega_4^{Pin^+} \xrightarrow{\chi} \mathbb{C}^\times.$$

A semistable bordism stabilizes to a Pin$^+$ bordism, so s is well-defined and respects disjoint union. Write $\omega = \chi \circ s$. This is a character by Lemma 3.1. Apply Input E over the algebraically closed field $\mathbb{C}$, whose characteristic is zero. It supplies an invertible symmetric monoidal functor

$$Z_\omega : Bord_4^{BPin^+(4),\beta} \to sVect_{\mathbb{C}},$$

landing in the Picard groupoid of super lines and their isomorphisms. For every closed (M,p) its value is $Z_\omega(M,p) = \omega\left([M,p]_{T\beta}\right) = \chi\left([M,p]_{Pin^+}\right) = I(M,p)$. The state-space assignments, gluing, and

monoidal coherence are supplied by the cited classification theorem. This is an application of that theorem, not an independent proof of it or a construction of an extended finite local categorical model. □

Theorem 3.6 (Pin⁺ detection as a consequence of established results). Using Inputs B and E, there exists an invertible four-dimensional Pin⁺ super TQFT whose closed values distinguish standard projective four-space from the Cappell–Shaneson exotic projective four-space, for every pairing of their Pin⁺ structures. Its closed values lie in the finite group of sixteenth roots of unity. This establishes the Pin⁺ detection statement, but does not by itself establish a specified finite categorical presentation with the required coherent antilinear time-reversal action.

Proof. Lemma 3.5 supplies the TQFT. Lemma 3.3 proves the separation, and Input B supplies homeomorphism of the underlying manifolds. Lemma 3.1 identifies the finite image. No lemma above specifies an extended categorical target, finite generators and relations for a local construction, or an antilinear action on the functor. Therefore none of those additional structures is a proved conclusion of this argument. The positive theorem has exactly the stated scope. □

4 What this does and does not settle

Lemma 4.1 (An oriented obstruction cannot be applied unchanged). A theorem quantified over oriented four-manifolds does not, without an additional argument, rule out detection of the pair R, Q above.

Proof. Lemma 2.5 proves R is non-orientable. Since orientability is preserved by homeomorphism, so is Q. Thus this pair is outside that theorem's domain. A universally quantified conclusion only applies to objects satisfying its hypotheses. □

Reutter's original no-go result is stated for oriented theories [2]. The later work [3, §1.1] states that semisimple 4-dimensional field theories can sometimes distinguish unoriented smooth structure, citing exactly the pair $\mathbb{R}P^4$ and the Cappell–Shaneson fake $\mathbb{R}P^4$, which are homeomorphic but not stably diffeomorphic. Two statements in the supplied announcement therefore need an orientation qualification: "Reutter proved that semisimple 4D TQFTs cannot detect exotic smooth structure" and "No TQFT built from finite algebraic data is known to detect exotic smooth structure" are accurate only for oriented manifolds. Likewise the genesis-script summary that "every semisimple-type invariant … agrees" on $\mathbb{R}P^4$ and the Cappell–Shaneson $\mathbb{R}P^4$ can only refer to the finite list the script computes (Lemma A.5); the Pin⁺ bordism character of Lemma 3.2 does not agree. Stable-diffeomorphism invariance and homeomorphism invariance are different assertions; the former must not silently be substituted for the latter in the Pin setting.

Lemma 4.2 (What the Pin⁺ obstruction does not show). The equalities $w_2 = 0$ and $w_2 + w_1^2 \neq 0$ do not construct a TQFT or certify its finite categorical origin.

Proof. Their proof supplies transition-function lifts or obstructions to such lifts. A TQFT additionally assigns state spaces to closed three-manifolds and maps to bordisms and satisfies composition and monoidal coherence. None of those assignments occurs in the lifting calculation. A claim of existence of these assignments therefore requires an additional construction or theorem, as in Lemma 3.5. A categorical presentation requires still further data not present in that lemma. □

Lemma 4.3 (Three meanings of finiteness). Finite image of closed partition functions and finite-dimensional state spaces do not, by definition alone, verify a finite local categorical presentation.

Proof. The first property describes a set of scalars; the second describes vector spaces after a functor has already been constructed. In the stronger local-presentation sense at issue here, a categorical presentation must specify a target, generating objects and morphisms, relations, duality data, and the compatibilities used in cutting and gluing. Neither a scalar set nor a bound on vector-space dimensions supplies

those data. Consequently the former checks cannot serve as a proof of the latter without an additional realization result. This is a statement about what has been established, not a claimed impossibility of such a realization. □

Lemma 4.4 (A scalar conjugation check is insufficient). For $k \in \mathbb{Z}/16$, $\overline{\chi(k)} = \chi(-k)$. This identity alone does not define T on TQFT state spaces.

Proof. Complex conjugation sends $e^{2\pi ik/16}$ to $e^{-2\pi ik/16}$. But a state-space action requires maps on the vector spaces assigned to three-manifolds and compatibility with all bordism maps. The displayed equality refers only to closed four-manifolds and specifies none of these maps. Thus it is a necessary sort of scalar compatibility in an appropriate model, not a construction of the action. □

The standard physical correspondence relates Pin$^+$ spacetime structure to time reversal squaring to fermion parity [5]. Invoking that correspondence is not a substitute for giving a coherent antilinear structure in a declared categorical model. In particular, the Euclidean Clifford lift of a reflection and the antiunitary Lorentzian time-reversal operator must not be identified without accounting for the conventions.

Lemma 4.5 (The theory of Theorem 3.6 cannot see homotopy 4-spheres). Let Σ be a smooth homotopy 4-sphere. Then Σ carries a unique Pin$^+$ structure and $Z_\omega(\Sigma) = 1 = Z_\omega(S^4)$. In particular no character of $\Omega_4^{Pin^+}$ bears on the smooth four-dimensional Poincaré conjecture.

Proof. Σ is simply connected, so $H^1(\Sigma;\mathbb{F}_2) = 0$ and $w_1 = 0$; $H^2(\Sigma;\mathbb{F}_2) = 0$, so $w_2 = 0$. Lemma 2.4 and the torsor argument of Lemma 2.6 give exactly one Pin$^+$ structure, and it is induced from the unique spin structure (with either orientation) by the homomorphism $Spin(4) \to Pin^+(4)$. Since $H^2(\Sigma;\mathbb{Z}) = 0$, the signature of Σ is 0. The spin bordism group $\Omega_4^{Spin} \cong \mathbb{Z}$ is detected by the signature (Rokhlin's theorem, the generator being the K3 surface of signature -16), so Σ is spin null-bordant and hence Pin$^+$ null-bordant. Thus $[\Sigma] = 0$ in $\Omega_4^{Pin^+}$ and $Z_\omega(\Sigma) = \chi(s[\Sigma]) = \chi(0) = 1$, by Lemma 3.5. The same argument applies to S^4. □

Disposition of the three entry points

Source entry point	Result of this paper
Semisimple obstruction on Pin manifolds	Answered in the negative in the literature for unoriented theories: [3, §1.1] states that semisimple unoriented 4D theories distinguish $\mathbb{R}P^4$ from the Cappell–Shaneson $\mathbb{R}P^4$. Theorem 3.6 supplies an explicit invertible Pin$^+$ example; whether it satisfies the precise semisimplicity definition of [2] is not re-verified here. No new general classification of Pin theories is proved.
Detect the Cappell–Shaneson example	Theorem 3.6 proves Pin$^+$ super-TQFT detection using identified external inputs. The full additional symmetry/presentation requirement is not verified.
Define finite categorical data	A definition must name the target, extension level, finite presentation or local state-sum data, and symmetry coherence. This paper does not replace that missing definition with a weaker one.

A complete solution in a stricter intended class must specify those choices, construct the theory there,

check gluing and time reversal, and evaluate it on the exotic pair. Alternatively, it must prove non-existence within a precisely specified class. Neither conclusion is claimed here. Nothing in the preceding theorem constructs or detects an exotic homotopy four-sphere; by Lemma 4.5 no theory factoring through $\Omega_4^{Pin^+}$ can do so, so the homotopy-sphere branch of Problem 5 needs genuinely non-invertible (and, by [2], non-semisimple) input.

5 Supplementary quaternionic exercises

These are complete solutions of Exercises 1, 5, and 10 in the supplied primer. They explain the algebra underlying the symmetry language; they are not additional solutions of the open existence question.

Exercise 1: the one-qubit antiunitary

Let K denote coordinate conjugation, $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, and $J_1 = YK$.

Lemma 5.1 (Square and conjugation formula). $J_1^2 = -I$, and $J_1 M J_1^{-1} = Y\overline{M}Y$ for every complex 2×2 matrix M.

Proof. For every vector v, $KYv = \overline{Y}\overline{v}$, so $KY = \overline{Y}K$. Since $\overline{Y} = -Y$ and $Y^2 = I$, $J_1^2 = Y\overline{Y} = -I$. Also $(YK)^{-1} = KY$ because $Y^2 = K^2 = I$. Hence $(YK)M(KY) = Y\overline{M}Y$. □

Lemma 5.2 (Fixed matrices). The complex matrices commuting with J_1 are precisely $\begin{pmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{pmatrix}$.

Proof. For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ direct multiplication gives $Y\overline{M}Y = \begin{pmatrix} \overline{d} & -\overline{c} \\ -\overline{b} & \overline{a} \end{pmatrix}$. Equality to M is equivalent to $d = \overline{a}$ and $b = -\overline{c}$. Put $\alpha = a$ and $\beta = c$. Conversely these conditions make every entry equal, proving both directions. □

Exercise 5: two quaternion factors

Lemma 5.3 (A real algebra homomorphism). The assignment $\Phi(a \otimes b)(x) = ax\overline{b}$ defines a unital real algebra homomorphism $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \to End_{\mathbb{R}}(\mathbb{H})$.

Proof. The formula is real-bilinear in a, b, so induces a map from the tensor product. Composition on pure tensors gives $a\left(cx\overline{d}\right)\overline{b} = acx\overline{bd}$, which is $\Phi(ac \otimes bd)(x)$. Bilinearity extends this equality to sums. Finally $\Phi(1 \otimes 1) = id$. □

Lemma 5.4 (Injectivity and surjectivity). Φ is an isomorphism; consequently $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong M_4(\mathbb{R})$.

Proof. Use the orthonormal basis $E = \{1, i, j, k\}$ of $\mathbb{H}$ with inner product $Re(\overline{x}y)$. Write $L_a(x) = ax$ and $R_b(x) = xb$. Expanding in E gives $tr(L_u R_v) = 4Re(u)Re(v)$: by bilinearity it suffices to take $u, v \in E$; the pair $(1, 1)$ gives trace four, a single imaginary factor has trace zero, and two imaginary factors have trace zero (diagonal signs sum to zero if the factors coincide and the diagonal vanishes otherwise).

For $a, b, c, d \in E$, the adjoint of $\Phi(a \otimes b)$ is $L_{\overline{a}}R_b$. Since right multiplications compose in reverse order,

$$tr\left(\Phi(a \otimes b)^* \Phi(c \otimes d)\right) = 4Re\left(\overline{a}c\right) Re\left(\overline{d}b\right) = 4\delta_{ac}\delta_{bd}.$$

The sixteen images are therefore orthogonal nonzero endomorphisms and linearly independent. Domain and codomain both have real dimension sixteen. Thus Φ is bijective. Choosing the basis E identifies the codomain with $M_4(\mathbb{R})$. □

Lemma 5.5 (Brauer order and tensor parity). $[\mathbb{H}]$ has order two in $Br(\mathbb{R})$, and $J_n = (Y^{\otimes n})K$ satisfies $J_n^2 = (-1)^n I$.

Proof. The previous lemma implies $[\mathbb{H}]^2 = [M_4(\mathbb{R})] = [\mathbb{R}]$. The class is nontrivial: the division algebra representative of the real matrix-algebra class is $\mathbb{R}$, whereas $\mathbb{H}$ is a noncommutative division algebra. Equivalently a split real central simple algebra of dimension four is $M_2(\mathbb{R})$, which has nonzero zero divisors, unlike $\mathbb{H}$. Thus the order is exactly two.

For the second assertion, $\overline{Y^{\otimes n}} = (-1)^n Y^{\otimes n}$ and $(Y^{\otimes n})^2 = I$, so $J_n^2 = Y^{\otimes n}\overline{Y^{\otimes n}} = (-1)^n I$. Pairing quaternion factors gives matrix algebras over $\mathbb{R}$ for even n and over $\mathbb{H}$ for odd n. The real/quaternionic parity therefore agrees with the sign of the antiunitary square. This is an algebraic correspondence, not a construction of a manifold invariant. □

Exercise 10: degeneracy and the ground multiplet

Lemma 5.6 (Kramers degeneracy). If $H = H^*$ on a finite-dimensional complex Hilbert space commutes with an antiunitary J satisfying $J^2 = -I$, every eigenspace of H has even complex dimension.

Proof. If $Hv = Ev$, then $E \in \mathbb{R}$ and $H(Jv) = J(Hv) = EJv$. To prove orthogonality use the convention that the inner product is conjugate-linear in its first entry. Antiunitarity gives $\langle Jx, Jy\rangle = \langle y, x\rangle$. Set $x = v, y = Jv$ to obtain $\langle Jv, -v\rangle = \langle Jv, v\rangle$, hence $\langle Jv, v\rangle = 0$. A unit eigenvector and its image thus span a two-dimensional J-invariant plane. Its orthogonal complement in the eigenspace is also J-invariant: antiunitarity preserves orthogonality, and J carries the plane to itself. Induction decomposes the eigenspace into such planes. Its dimension is even. □

Lemma 5.7 (Which energy gap is meaningful). On the finite-dimensional n-qubit Hilbert space, for odd n, a self-adjoint J_n-symmetric Hamiltonian has a degenerate ground eigenspace. If an excited eigenvalue exists, the gap above the ground multiplet is $\min\{E - E_0 : E \in spec(H), E > E_0\}$.

Proof. The tensor-square calculation above gives $J_n^2 = -I$ for odd n. Kramers degeneracy supplies at least two orthogonal ground vectors. Ordering energies with multiplicity therefore gives $E_1 - E_0 = 0$, irrespective of whether the next distinct energy is positive distance away. Removing the entire ground eigenspace leaves exactly the distinct energies $E > E_0$, whose smallest difference is the displayed expression. This explains the source's convention for Problem 1. □

A The exact motivating exercise and the derivation of the question

Exact source exercise

The following is Exercise 9 from Part I, Section 5, printed page 9 of the supplied PDF. Its difficulty rating is three stars; mathematical typesetting is normalized, with the wording preserved:

> **Exercise 9.** Using Stiefel–Whitney classes, show that $\mathbb{RP}^4$ is non-orientable, admits Pin$^+$ structures and admits no Pin$^-$ structure.

The Problem 5 reading path explicitly refers to this exercise. Section 6's "5 · Exotic detection" description also singles out the Stiefel–Whitney calculation of projective space. This establishes the documentary

connection. It does not establish the authors' private historical thought process, nor make the conjecture a logical consequence of the exercise.

Lemma A.1 (The exercise selects the positive Pin sign). The result of Exercise 9 makes $\mathbb{R}P^4$ an admissible closed four-manifold for evaluation by a Pin$^+$ theory and excludes it from a Pin$^-$-only theory. By Lemma 2.7 the same holds for the Cappell–Shaneson $\mathbb{R}P^4$.

Proof. In the ordinary four-dimensional bordism category, objects are closed three-manifolds, while a closed four-manifold is a morphism from the empty object to itself. Such a structured four-manifold must carry the specified tangent lift. Lemma 2.5 supplies precisely the positive lift and proves the negative obstruction nonzero, and Lemma 2.7 transfers both statements to any homotopy $\mathbb{R}P^4$. Therefore, among Pin± theories, the proposed projective-space test must use Pin$^+$ if it is to include the standard manifold. (Theories with other tangential structures, for example unoriented or Pin^c theories, are not excluded by this argument.) □

Lemma A.2 (The symmetry motivation is compatible at the level of signs). Let T be an antilinear operator on a complex super vector space, and let P be its fermion-parity operator. If $T^2 = P$, then T preserves the grading and restricts to a real structure on the even sector and a quaternionic structure on the odd sector; the latter agrees with $J_n^2 = -I$ for odd n.

Proof. First, associativity gives $TP = TT^2 = T^3 = T^2T = PT$. If $Pv = \lambda v$ with $\lambda = \pm 1$, then $P(Tv) = T(Pv) = \lambda Tv = \lambda Tv$, because λ is real; so T preserves each parity eigenspace. On the even subspace $P = I$, so $T^2 = I$. On the odd subspace $P = -I$, so $T^2 = -I$. A complex vector space with an antilinear map squaring to $-I$ has a quaternion action: multiplication by i is one generator and T is a second; antilinearity gives $Ti = -iT$, both squares are $-I$, and $(iT)^2 = i(Ti)T = i(-iT)T = T^2 = -I$, so i, T, iT satisfy Hamilton's relations. Section 5's calculation gives exactly this sign for J_n at odd n. This proves compatibility of the symmetry algebra; it does not identify a particular TQFT realization. □

Lemma A.3 (Adding an exotic test pair is an extra input). Exercise 9 alone does not assert existence of an exotic partner for $\mathbb{RP}^4$; Input B supplies that additional premise.

Proof. Every assertion of Exercise 9 concerns the tangent lift of the single standard smooth manifold $\mathbb{RP}^4$. It contains neither a second manifold nor a homeomorphism or a non-diffeomorphism claim. Input B supplies Q and its homeomorphism to R. Lemma 3.3 then detects their smooth distinction using further bordism information. The second manifold is therefore an additional ingredient, not an output of the characteristic-class exercise. □

Lemma A.4 (From a test object to an existence question). The exercise, the symmetry sign, and an exotic pair motivate, but do not logically prove, the existence assertion in Problem 5.

Proof. The exercise produces an admissible structured object. The sign calculation specifies a candidate symmetry type. The exotic pair specifies a desired test. An existence proof must additionally give a theory in the requested class and verify unequal evaluations. None of the first three items assigns a vector space or a bordism map, as Lemma 4.2 explains. Their combination therefore identifies what to search for, rather than supplying the missing witness. Section 3 gives such a witness for the stated super-TQFT version using Input E, while Section 4 records the extra conditions not verified there. □

Lemma A.5 (The finite numerical genesis script cannot close the gap). Agreement of a chosen finite list of invariants on an exotic pair does not prove agreement for every TQFT in Problem 5.

Proof. A list checks finitely many functions $f_1, \dots, f_m$. The conclusion is only $f_i(R) = f_i(Q)$ for the tested indices. To infer agreement for every member of a larger class requires a completeness theorem identifying that class with the tested functions or proving all its functions are determined by them. No such completeness theorem is part of the listed computations. Indeed Lemma 3.3 exhibits additional

Pin^+ bordism information separating the pair. The script's own disclaimer that it does not prove or disprove the challenge is therefore essential. □

B Verification and remaining work

Lemma B.1 (Internal arithmetic checks). The characteristic-class expansion, the exponent comparison, and the averaged detector are mutually consistent.

Proof. Modulo two, the binomial coefficients in $(1+a)^5$ are $(1,1,0,0,1,1)$, with the last term removed by $a^5 = 0$. Modulo sixteen, $-1 = 15$ and $-9 = 7$; neither 7 nor 9 equals 1 or 15. Finally adding eight to an exponent multiplies its root of unity by -1, agreeing with the opposite averages in Lemma 3.4. These elementary checks verify the local computations, not the external bordism or functor-classification theorems. □

This edition accepts the elementary mathematics and supplies the exercise proof, its extension to every homotopy $\mathbb{R}P^4$ (Lemma 2.7), and the explicitly delimited, literature-dependent Theorem 3.6 together with its limitation on homotopy spheres (Lemma 4.5). Its repairs make the Pin obstruction cocycle and its universal identification, the stabilization map, antilinearity hypothesis, grading preservation, and bordism-category terminology explicit, and they correct the citations for Inputs B and E. It does not supply an independent proof of Inputs B and E, an explicit finite categorical local model, or a coherent antilinear T for that model. Consequently it would be incorrect to submit this document as a completely self-contained solution of every condition in the original request. A subsequent complete solution must close those named gaps and check the authoritative problem statement.

C Assessment record (second review, 3 November 2026)

Every lemma, theorem and proof was checked by hand: each computation was redone and each logical step re-derived. External citations were checked against the sources that could be reached (arXiv:2206.10031 v3, Stolz [6] and Debray [7]). The publication details of [2] and [3] and the precise hypotheses of [3, Theorem 3.14] could not be fully confirmed from the accessible text and should be checked before submission.

Item	Verdict	What was checked or changed
Lemma 1.1	Correct	No change.
Lemma 2.1	Correct	Cellular boundaries 0 or 2; bar-complex cup powers u^k evaluate to 1 on the cycle [g|…|g]; restriction to the 4-skeleton is an isomorphism through degree 4. No change.
Lemma 2.2	Correct; wording fixed	"Middle endomorphism line" replaced by the summand $\mathrm{Hom}(\gamma,\gamma) \cong \varepsilon$; final step made explicit.
Lemma 2.3	Correct	$(1+a)^5 \equiv 1 + a + a^4 \bmod (2, a^5)$. $w_4 = a^4$ agrees with $\chi(\mathbb{R}P^4) = 1 \bmod 2$.
Lemma 2.4	Correct statement; proof repaired	Cocycle c(g,h) re-derived and correct. The proof mixed a flag-bundle splitting argument with a group-cohomology computation; replaced by the universal class $o^\pm \in H^2(BO(r))$, injectivity of restriction to $(\mathbb{Z}/2)^r$, and naturality. Pin± sign convention stated (Kirby–Taylor).
Lemma 2.5	Correct	No change.
Lemma 2.6	Correct	H^1-torsor count gives 2. No change.

Item	Verdict	What was checked or changed
Lemma 2.7	Added	The draft used “their two Pin⁺ structures” for Q without proof. Now proved from homotopy invariance of Stiefel–Whitney classes (Wu).
Input B	Citation corrected	The ±1 / ±9 values were attributed to [3, §6.2.2], which could not be confirmed. Re-sourced to Stolz [6] as stated in Debray [7, Ex. 3.4]; ℤ/16 to Kirby–Taylor [8]; homeomorphism to Hambleton–Kreck–Teichner [9] and [3, §1.1].
Input E	Citation corrected	The classification of invertible super theories by characters of the semistable bordism group is Theorem 3.14 in arXiv v3 of [3], not 3.15.
Lemma 3.1	Correct	No change.
Lemma 3.2	Correct	No change.
Lemma 3.3	Correct	{1,15} ∩ {7,9} = ∅. Also checked that the separation survives every automorphism of ℤ/16 ($9u \not\equiv \pm u \bmod 16$ for odd u).
Lemma 3.4	Correct, overstated	The values $\pm\cos(\pi/8)$ depend on the chosen identification; restated as $\pm\cos(\pi u/8)$, u odd, with non-vanishing proved.
Lemma 3.5	Correct given Input E	Factorisation $\omega = \chi \circ s$ checked; citation number fixed via Input E.
Theorem 3.6	Correct given Inputs B and E	Scope statement accurate. No change.
Lemma 4.1	Correct	Surrounding paragraph corrected: [3] is cited for what its §1.1 states; the announcement’s unqualified claims about Reutter’s theorem and about finite algebraic data are flagged.
Lemmas 4.2–4.4	Correct	Scope observations; no change.
Lemma 4.5	Added	Bordism-character theories take the value 1 on every homotopy 4-sphere (Rokhlin: $\Omega_4(\mathrm{Spin}) \cong \mathbb{Z}$ detected by signature).
Entry point 1	Disposition corrected	The draft said only that the oriented-to-Pin inference is invalid. The literature ([3, §1.1]) already answers the question negatively for unoriented theories.
Lemmas 5.1–5.2	Correct	$Y \bar{M} Y = (đ, -\bar{c}; -\bar{b}, \bar{a})$ recomputed.
Lemmas 5.3–5.4	Correct	$\mathrm{tr}(L_u R_v) = 4\mathrm{Re}(u)\mathrm{Re}(v)$ checked on all basis pairs; adjoint and trace identity checked.
Lemma 5.5	Correct	The final sentence is a heuristic correspondence, as the text already says.
Lemmas 5.6–5.7	Correct	$\langle Jv,v\rangle = 0$ and J-invariance of orthogonal complements checked.
Lemma A.1	Correct, overstated	“Must use Pin⁺” restricted to Pin± theories; extended to the Cappell–Shaneson ℝP⁴ via Lemma 2.7.
Lemma A.2	Correct; detail added	Eigenspace preservation and $(iT)^2 = -I$ written out.
Lemmas A.3–A.5, B.1	Correct	No change.

References

[1] R. Cooperman and A.E. Petrunina V.V. *The Cooperman–Petrunina Challenge Series: Announcement, Primer and Reading Guide*. Supplied PDF, Part I (2026), especially pp. 8–10, 13. DOI recorded in the supplied file: https://doi.org/10.5281/zenodo.23423138. Source text CC BY 4.0, https://creativecommons.org/licenses/by/4.0/. The accompanying Part II is the underlying algebra paper, not the separate five-problem deposit.

[2] D. Reutter. Semisimple four-dimensional topological field theories cannot detect exotic smooth structure. *Journal of Topology* 16 (2023), 542–566. https://doi.org/10.1112/topo.12288. Scope: oriented theories.

[3] D. Reutter and C. Schommer-Pries. *Semisimple Field Theories Detect Stable Diffeomorphism.* arXiv:2206.10031 (v3), especially §1.1 and Theorem 3.14. Journal details as given in the earlier draft (*Advances in Mathematics* 482 (2025), 110598) not independently confirmed. External source for the functorial extension; these results are not newly proved here.

[4] W. J. Oledzki. On the eta-invariant of Pin^+-operator on some exotic 4-dimensional projective space. *Compositio Mathematica* 78 (1991), 1–27. https://www.numdam.org/item/CM_1991__78_1_1_0.pdf.

[5] A. Kapustin, R. Thorngren, A. Turzillo and Z. Wang. Fermionic symmetry protected topological phases and cobordisms. *JHEP* 12 (2015), 052. Sections 2–3 discuss Pin lifts and the time-reversal convention. https://arxiv.org/abs/1406.7329.

[6] S. Stolz. Exotic structures on 4-manifolds detected by spectral invariants. *Inventiones Mathematicae* 94 (1988), 147–162.

[7] A. Debray. Stable diffeomorphism classification of some unorientable 4-manifolds. arXiv:2102.03965 (2021), Example 3.4.

[8] R. C. Kirby and L. R. Taylor. Pin structures on low-dimensional manifolds. In *Geometry of Low-Dimensional Manifolds 2*, LMS Lecture Note Series 151, Cambridge University Press (1990), 177–242.

[9] I. Hambleton, M. Kreck and P. Teichner. Nonorientable 4-manifolds with fundamental group of order 2. *Transactions of the AMS* 344 (1994), 649–665.

[10] S. E. Cappell and J. L. Shaneson. Some new four-manifolds. *Annals of Mathematics* 104 (1976), 61–72.

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